# Separation and Depression in Infant Gorillas

MICHAEL P. HOFF
*Dalton College*
*Dalton, Georgia*
*and*
*Zoo Atlanta*
*Atlanta, Georgia*

RONALD D. NADLER
*Yerkes Regional Primate Research Center*
*Emory University*
*Atlanta, Georgia*

KATHRYN T. HOFF
*Dalton Public Schools*
*Dalton, Georgia*

TERRY L. MAPLE
*Georgia Institute of Technology*
*and*
*Zoo Atlanta*
*Atlanta, Georgia*

Three 27-month-old infant gorillas living with their mothers and a silverbacked male were separated to a cage for 24 weeks. The infants initially showed threat responses and increased locomotion, characteristic of the protest stage of anaclitic depression in children. Within several days, these were replaced by dorso–ventral contact among the infants as well as self-holding and fetal positioning. Additionally, social and solitary play and object examination occurred at lower levels through separation than in the pre-separation condition. These changes were characteristic of the despair stage of separation. There was a substantial recovery of many infant nonsocial and social behaviors in the later months of the separation. Upon reunion, the infants did not immediately engage in attachment behaviors with their mothers, and spent more time in contact with each other than with their mothers for the first several days, indicating detachment. Following this, there was an increase in mother–infant attachment behaviors. © 1994 John Wiley & Sons, Inc.

Reprint requests should be sent to Michael P. Hoff, Ph.D., BA/SS, Dalton College, Dalton, GA 30720, U.S.A.

Received for publication 18 March 1994
Revised for publication 3 June 1994
Accepted for publication 10 June 1994

*Developmental Psychobiology* 27(7):439–452 (1994)
© 1994 by John Wiley & Sons, Inc.
CCC 0012-1630/94/070439-14

Throughout this century, the value of animal models of human psychopathology has been recognized. The value lies in the ability to produce, study, and cure various pathologies in animals, and apply the research findings to humans. Since Pavlov (1928) presented his model of experimental neurosis, psychopathology has been produced experimentally in a variety of species. One area that has been extremely productive has been the induction of experimental depression in nonhuman, usually primate, subjects.

The early studies of primate models of separation generally separated infants from their mothers, objects of significant attachment to the infants (Akiskal & McKinney, 1975), although separation of infants from other objects of attachment [e.g., inanimate surrogates (Novak, 1973) and peers (Erwin, Brandt, & Mitchell, 1973; Rasmussen & Reite, 1982; Suomi, Harlow, & Domek, 1970)] has also produced depression in monkeys.

The mother–infant separation studies in rhesus macaques and pigtail macaques have typically found a biphasic response reminiscent of that reported in humans by Spitz (1946) and Bowlby (1960, 1969, 1973, 1980) of protest upon initial separation and despair as the separation continued (Hinde, 1972; Hinde & Spencer-Booth, 1971; Kaufman & Rosenblum, 1967; Jensen & Tolman, 1962; Seay, Hansen, & Harlow, 1962; Spencer-Booth & Hinde, 1967, 1971a, 1971b), although monkeys typically do not show the final stage of detachment that is shown by children. The biphasic response of protest and despair in monkeys is not invariant, however. An early study by Rosenblum and Kaufman (1968) found that although bonnet macaque infants showed a brief period of agitation upon separation, they did not exhibit depression. Other researchers have found little or no evidence of separation depression in other primates, including patas monkeys and squirrel monkeys (Jones & Clark, 1973; Kaplan, 1970; Preston, Baker, & Seay, 1970).

Although there is a long history of separation studies in monkeys, there are relatively few reports of mother–infant separation in great apes. Goodall (1990; Lawick-Goodall, 1971) has provided several case reports of the effects of maternal loss in young chimpanzees, some nutritionally dependent and others nutritionally independent of their mothers. In the younger animals, maternal loss resulted in death. In older infants and juveniles, a variety of responses were found, ranging from relatively brief depressions (several weeks) to the death of the orphaned animal following a decline in social and self-maintenance behavior. Experimental work with the great apes has been reported by Nadler and colleagues (Bard & Nadler, 1983; Codner & Nadler, 1984; Nadler & Codner, 1983), who separated infants from mothers or peers among all three species of great apes. In the mother–infant separations (Codner & Nadler, 1984; Nadler & Codner, 1983), there was evidence of the complete sequence of protest, despair, and detachment responses, as found in children. These studies, in which a detachment response was evident are consistent with the report of Plimpton and Rosenblum (1981), who found a detachment response in infant pigtail macaques upon reunion when the mothers were prevented from retrieving the infants.

The present study examines the effects of a 24-week separation of 3 group-living infant lowland gorillas (*Gorilla gorilla gorilla*) from their mothers. Infants were removed from the group compound and housed together. Social housing was expected to prevent debilitating depression (see Suomi, Eisele, Grady, & Harlow, 1975 for data regarding the effects of the separation social situation on depression). The 24-week length of the separation was expected to allow despair to recover (see Kaufman & Rosenblum, 1967) and, additionally, to be long enough for detachment to develop.

This separation study was not done for the sole purpose of testing for the possibility of behavioral depression in infant gorillas. Rather, a veterinary/husbandry decision was

10982302, 1994, 7, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/dev.420270703 by Universitaetsbibliothek, Wiley Online Library on [01/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Table 1
*Subject Information*

| Name | Identification | Parentage |
|---|---|---|
| Rann | Adult M | Wild born |
| Choomba | Adult F | Wild born |
| Machi | Infant F | Rann or Calabar × Choomba |
| Segou | Adult F | Wild born |
| Bom Bom | Infant M | Rann × Segou |
| Shamba | Adult F | Wild born |
| Akbar | Infant M | Rann or Calabar × Shamba |

made to remove the infants from the mothers for a period of time to allow lactation to cease and the females to resume cycling, become fertile and, hopefully, more infants to be born. This study was opportunistic, therefore, given the decision to separate the mothers and infants.

## Method

### Subjects

The subjects were 3 mother–infant pairs of lowland gorillas living in a social group with a silverbacked male (Table 1) in an outdoor compound approximately 23 m × 30 m in size at the Yerkes Regional Primate Research Center Field Station. The group had been maintained intact for 3 years as the study began except for a 6-week experimental removal of the adult male when the infants were 18 months old. The study began in May, 1978[1] when the infants were 27 months old, within Schaller's (1963) and Napier and Napier's (1967) 36-month cutoff age of infancy. The history and housing of this group are described in detail in Nadler (1976) and Tilford and Nadler (1978).

The physical separation of mothers and infants took place outside the compound with a minimum of handling and trauma by immobilizing the mothers individually with 5 mg/kg ketamine hydrochloride. The infants were removed from the mothers approximately 20 min later. The female infant (Machi) was also injected with ketamine hydrochloride due to her extreme agitation upon separation. The infants were moved to a housing area approximately 100 yd from the compound, leaving the 3 adult females and the adult male in the compound. The male infants were released simultaneously into a two-compartment cage consisting of an indoor compartment approximately 2.3 m × 2.3 m × 2 m high and an outdoor compartment 2.3 m × 3.7 m × 2.3 m high, separated by a guillotine door. The female infant was introduced to the males upon full recovery from the effects of the drug. The infants were separated from their mothers for 24 weeks, at which time they were reintroduced simultaneously into the home compound containing the adults.

Observations specific to this study began 3 weeks before separation and continued for 4 weeks after reunion. Observations took place 3–4 days per week in the morning, a time of maximal activity in both the lowland gorilla (Jones & Sabater Pí, 1971) and mountain gorilla (Schaller, 1963). Observations were made using a sequential activity record, scoring all solitary and social behaviors of interest (Table 2) in the focal animal. Data were collected during 20-min time periods. Observations were made in full view

10982302, 1994, 7, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/dev.420270703 by Universitaetsbibliothek, Wiley Online Library on [01/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Table 2
*Behavioral Definitions*

| |
|---|
| Ventro–ventral—Animals' ventral surfaces in contact. |
| Dorso–ventral—One animal's ventrum pressed against another animal's dorsum. Herein used exclusively in "back ride" position (infant lying on mother's back). |
| Other contact—Any form of contact excluding ventro–ventral and dorso–ventral. |
| Proximate—Animals not in contact, but within 1 m of another. |
| Distant—Animals between 1 and 5 m apart. |
| Far Distant—Animals beyond 5 m apart. |
| Follow—Moving behind and in the same direction as another animal. |
| Nipple contact—Animal's mouth placed on another's nipple with or without suckling. |
| Social examination—Inspect another animal with fingers, feet, and/or mouth (includes stare). |
| Active social play—Social play with whole body involvement, including wrestling, rolling, chasing, etc. |
| Moderate social play—Social play (see above) with less than whole body involvement, often gentle rocking, mouthing, or tugging. |
| Hold other—One animal holding another in both arms. |
| Sit—Common usage. |
| Lie—Common usage. (SI and LI herein combined into Sit/lie). |
| Self-hold—Animal not moving with arms wrapped around self. |
| Fetal—Animal not moving, with legs drawn up to chest, arms wrapped around self, and head down. (SH and FE herein combined into Self-hold/fetal). |

of the subjects, from an observation deck overlooking the gorilla compound prior to separation and following reunion, and directly in front of the two-compartment separation cage during separation. The infants were locked into the outdoor compartment of the separation cage during data collection. Order of observation was randomly determined.

Data were tabulated on summary sheets and averaged over 4-week blocks (hereafter referred to as "months"). Because of the small number of subjects, descriptive statistics were used. Mother–infant, infant–infant, and infant nonsocial behaviors are presented over: (a) preseparation, (b) separation (subdivided into 6 months), and (c) postseparation periods. Data are presented in mean minutes per 20-min observation period. Mean individual scores are presented in Tables 3–5 to allow examination of individual differences. Overall infant means for several behaviors are also presented graphically.

## Results

### Preseparation

The infants at 27 months were all nutritionally independent of their mothers (see Hoff, Nadler, & Maple, 1981a). Preseparation mother–infant spatial proximity variables (Table 3) show that the infants were spending little time in contact with their mothers (Figure 1), and were proximate an average of only 0.1 min per observation. Nipple contact was observed in only 1 infant. Likewise, social examination had almost completely disappeared by 27 months of age.

In the preseparation period, infants interacted with each other more than with their mothers. Most infant interactions (Table 4) were in the form of play, as shown by the calculation of play as a percentage of contact (mean for all infants of 97.2%). Infant social play was typically active social play (mean 3.5 min) rather than moderate social play (mean 1.3 min) in the preseparation period (see Hoff, Nadler, & Maple, 1981b). The infants were proximate to one another for a mean of 0.8 min per observation period. Infant–infant hug/hold was not observed.

Several infant nonsocial behaviors were measured (Table 5). In the preseparation

10982302, 1994, 7, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/dev.420270703 by Universitaetsbibliothek, Wiley Online Library on [01/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Table 3
*Individual Scores for Preseparation and Postseparation Mother–Infant Interactions in Mean Min Per 20-Min Observation*

| Behavior | | Preseparation | Postseparation |
|---|---|---|---|
| Ventro–ventral | A | 0.0 | 0.3 |
| | B | 0.0 | 0.5 |
| | M | 0.0 | 0.4 |
| Dorso–ventral | A | 0.0 | 0.8 |
| | B | 0.2 | 0.7 |
| | M | 0.0 | 2.4 |
| Other contact | A | 0.2 | 0.8 |
| | B | 0.4 | 1.6 |
| | M | 0.6 | 0.5 |
| Proximate | A | 0.1 | 1.7 |
| | B | 0.0 | 1.3 |
| | M | 0.2 | 0.9 |
| Distant | A | 1.0 | 3.9 |
| | B | 0.5 | 3.7 |
| | M | 2.4 | 2.3 |
| Far Distant | A | 18.7 | 12.4 |
| | B | 18.9 | 12.1 |
| | M | 16.7 | 13.4 |
| Follow | A | 0.1 | 0.5 |
| | B | 0.0 | 0.1 |
| | M | 0.1 | 0.7 |
| Nipple contact | A | 0.0 | 0.3 |
| | B | 0.1 | 0.4 |
| | M | 0.0 | 0.1 |
| Social exam | A | 0.0 | 0.5 |
| | B | 0.0 | 0.4 |
| | M | 0.1 | 0.2 |

*Note.* A = Akbar (male); B = Bom-Bom (male); M = Machi (female).

period, infants spent 49% of the time engaged in either object examination or solitary play (Figure 2). A majority of the time (mean of 13.1 min per observation) was spent sitting or lying still, the remainder either walking, running, climbing, etc. Infants exhibited no self-hold/fetal behavior in the preseparation period.

## Separation

### *Immediate Response*

The immediate response of all infants to separation was loud screaming and cough threats. The female infant responded so vigorously that she was given ketamine hydrochloride and placed in a transport box. The 2 males were placed in transport boxes out of sight of their mothers, where they continued to vocalize and huddled in the corners. The 3 mothers were still anesthetized at the time of separation, and thus did not respond.

10982302, 1994, 7, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/dev.420270703 by Universitaetsbibliothek, Wiley Online Library on [01/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

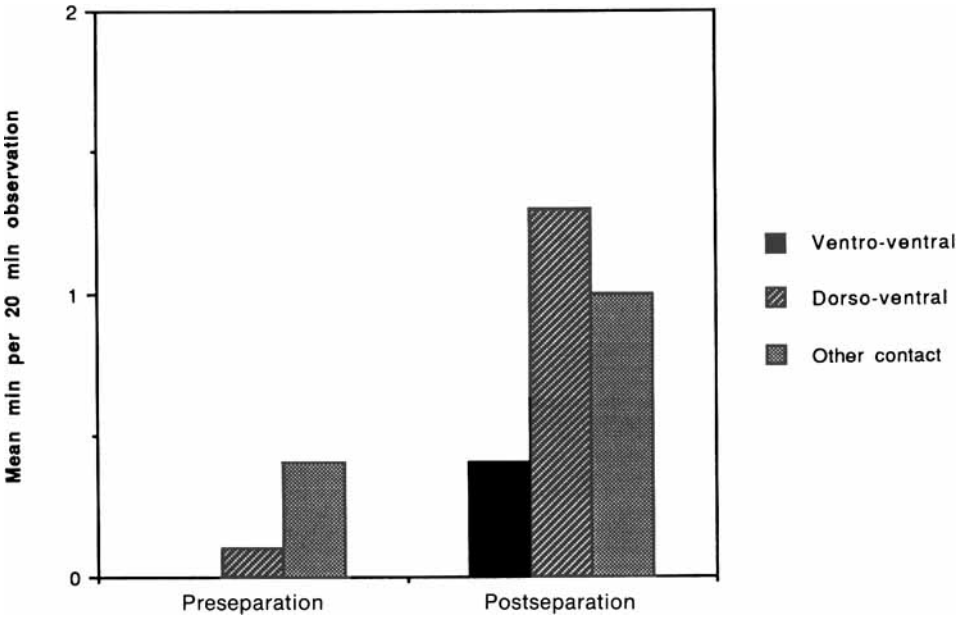



Fig. 1. Preseparation and postseparation mother–infant Ventro–Ventral, Dorso–Ventral and Other Contact in mean minutes per 20-min observation.

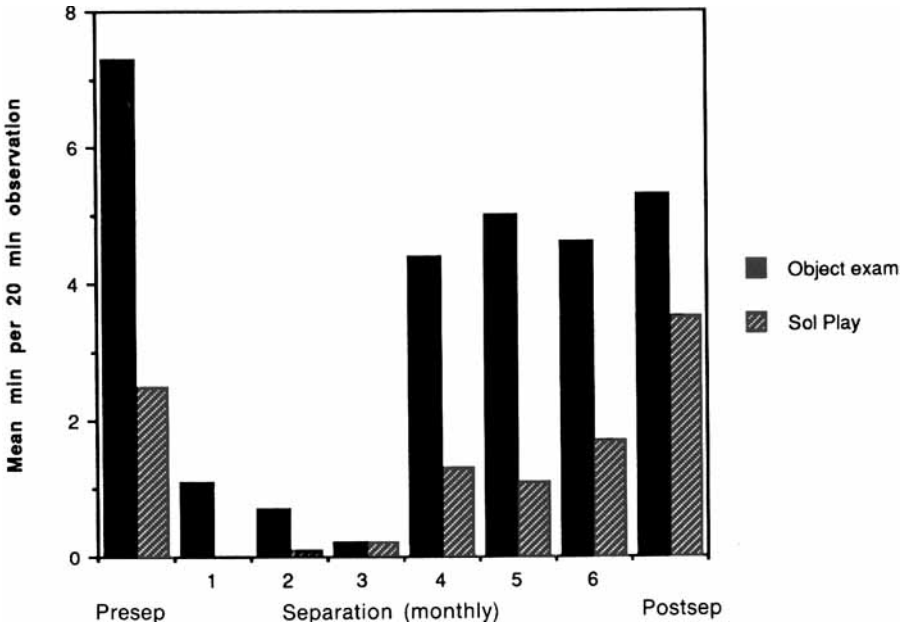



Fig. 2. Preseparation, separation (monthly), and postseparation infant Object Examination and Solitary Play in mean minutes per 20-min observation.

10982302, 1994, 7, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/dev.420270703 by Universitaetsbibliothek, Wiley Online Library on [01/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Table 4
*Individual Scores for Preseparation, Separation (Monthly), and Postseparation Infant–Infant Interactions in Mean Min Per 20-Min Observation*

| Behavior | | Preseparation | Separation 1 | 2 | 3 | 4 | 5 | 6 | Postseparation |
|---|---|---|---|---|---|---|---|---|---|
| Total contact | A | 5.1 | 13.8 | 10.8 | 13.9 | 12.9 | 10.4 | 6.6 | 2.5 |
| | B | 5.2 | 14.8 | 9.9 | 7.4 | 5.3 | 4.7 | 2.9 | 2.6 |
| | M | 4.0 | 16.5 | 14.4 | 16.3 | 7.9 | 10.1 | 5.1 | 1.6 |
| Proximate | A | 0.5 | 4.7 | 6.0 | 2.9 | 3.4 | 3.3 | 5.4 | 1.4 |
| | B | 0.8 | 4.3 | 5.4 | 1.4 | 3.8 | 8.5 | 3.5 | 1.9 |
| | M | 1.0 | 3.1 | 1.9 | 1.0 | 4.3 | 5.2 | 6.1 | 1.3 |
| Active play | A | 3.5 | 0.7 | 1.2 | 1.3 | 1.1 | 0.3 | 1.8 | 1.9 |
| | B | 4.2 | 0.8 | 1.0 | 0.1 | 1.0 | 0.4 | 1.2 | 1.4 |
| | M | 2.4 | 0.1 | 0.2 | 0.3 | 0.4 | 0.1 | 0.1 | 0.5 |
| Moderate play | A | 1.4 | 0.4 | 3.1 | 1.4 | 3.0 | 3.0 | 2.2 | 0.2 |
| | B | 0.9 | 0.2 | 2.3 | 2.3 | 1.3 | 2.7 | 1.0 | 0.5 |
| | M | 1.5 | 0.4 | 1.8 | 1.9 | 2.6 | 2.0 | 1.0 | 0.5 |
| Hold/hug | A | 0.0 | 3.6 | 1.8 | 2.4 | 0.5 | 5.0 | 0.7 | 0.0 |
| | B | 0.0 | 2.1 | 2.4 | 1.4 | 0.1 | 0.1 | 0.0 | 0.1 |
| | M | 0.0 | 4.3 | 2.5 | 3.0 | 0.6 | 2.7 | 1.3 | 0.0 |
| Play as % of contact | A | 96.1 | 7.9 | 39.8 | 19.4 | 31.8 | 31.7 | 60.6 | 84.0 |
| | B | 98.1 | 6.8 | 33.3 | 32.4 | 43.4 | 44.7 | 75.8 | 73.0 |
| | M | 97.5 | 3.0 | 13.9 | 13.5 | 38.0 | 30.7 | 21.6 | 62.5 |

*Note.* A = Akbar (male); B = Bom-Bom (male); M = Machi (female).

Infant vocalizations remained high for the first 2 days of separation, as did locomotion, although this is not reflected in a major drop in sit/lie behaviors in the 1st separation month summary. Also immediately apparent was an increase in infant–infant dorso–ventral contact, and infants holding one another. The mean occurrence of holding dropped slightly by the end of Day 2, with other contact, proximate, and infant self-hold/fetal positions appearing. There was no object examination nor any form of play for the first 2 days of separation. These results are indicative of an infant protest stage, grading into despair by the end of the 2nd day.

### *Course of Separation*

Following the period of agitation, behavioral depression ensued and continued through most of the 1st month of separation, with a subsequent substantial recovery. Most of the infant–infant behaviors (Table 4) and all of the infant nonsocial behaviors (Table 5) are suggestive of the despair and recovery stages of anaclitic depression.

Infant–infant contact showed a dramatic rise from a mean of 4.9 min preseparation to 15.1 min upon separation, then generally decreasing over the remaining separation months (Figure 3). One infant, Bom-Bom, recovered more quickly than the others, showing a decline of contact to approximate preseparation levels by the 4th month. Infants proximate to another also increased upon separation, from a mean of 0.8 min to 4.0 min in the 1st separation month. This behavior increased in the following month as contact declined somewhat; it was variable over the remaining separation months.

Infants holding one another was not observed in the preseparation condition. It occurred for 3.3 min per observation in the 1st separation month, declining in the

10982302, 1994, 7, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/dev.420270703 by Universitaetsbibliothek, Wiley Online Library on [01/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Table 5
*Individual Scores for Preseparation, Separation (Monthly), and Postseparation Infant Nonsocial Behaviors in Mean Min Per 20-Min Observation*

| Behavior | | Preseparation | Separation | | | | | | Postseparation |
|---|---|---|---|---|---|---|---|---|---|
| | | | 1 | 2 | 3 | 4 | 5 | 6 | |
| Sit/Lie | A | 12.8 | 16.5 | 9.9 | 17.5 | 15.4 | 17.3 | 15.2 | 7.6 |
| | B | 13.2 | 15.5 | 12.5 | 14.1 | 13.0 | 14.0 | 10.9 | 8.0 |
| | M | 13.3 | 17.4 | 15.1 | 14.3 | 14.3 | 15.6 | 15.8 | 8.8 |
| Object exam | A | 8.9 | 1.5 | 0.2 | 0.0 | 2.6 | 4.6 | 3.3 | 4.7 |
| | B | 4.7 | 1.1 | 1.0 | 0.4 | 5.4 | 6.5 | 4.4 | 5.7 |
| | M | 8.3 | 0.7 | 0.8 | 0.1 | 5.0 | 4.0 | 6.1 | 5.6 |
| Solitary play | A | 2.8 | 0.0 | 0.1 | 0.1 | 0.5 | 0.6 | 0.5 | 2.4 |
| | B | 2.5 | 0.0 | 0.1 | 0.4 | 2.1 | 2.2 | 2.4 | 4.3 |
| | M | 2.3 | 0.0 | 0.0 | 0.0 | 1.3 | 0.6 | 2.2 | 3.9 |
| Self-hold/fetal | A | 0.0 | 5.2 | 0.0 | 0.1 | 0.0 | 0.0 | 0.0 | 0.0 |
| | B | 0.0 | 3.6 | 0.3 | 0.8 | 0.0 | 0.0 | 0.0 | 0.0 |
| | M | 0.0 | 5.9 | 1.0 | 1.7 | 0.0 | 0.0 | 0.0 | 0.0 |

*Note.* A = Akbar (male); B = Bom-Bom (male); M = Machi (female).

following months, with a single substantial increase to a mean of 2.6 min in the 5th month of separation (Figure 4), largely due to a very high individual score (5.0 min) in a male infant.

Social play, as noted above, was categorized as active and moderate. Active social play decreased from a mean of 3.5 min preseparation to 0.5 min in the 1st separation month, and stayed at low levels through the remainder of the separation months. Moderate social play also declined, from a mean of 1.3 min (preseparation) to 0.3 min (1st month of separation) per observation period. Following, however, was a substantial increase to levels higher than in the preseparation condition through the remainder of the separation months. Play as a percentage of contact dropped precipitously upon separation, from an average of 97.2% preseparation to 5.9% in the 1st separation month. It fluctuated thereafter, with a general trend toward an increasing percentage of play, particularly among the males, when the infants were in contact.

Infant nonsocial behaviors included object examination, solitary play, and self-hold/fetal positioning. Self-hold/fetal (Figure 4) changed from 0.0 min preseparation to a mean of 4.9 min per observation in the 1st separation month, then rapidly declined and disappeared by the fourth separation month. Infant solitary play and object examination showed similar patterns of substantial declines from preseparation to 1st month separation with a level of recovery coming later in the months of separation (Figure 2).

## Postseparation

Infants were returned to the large compound simultaneously. Immediately prior to the infants' release from the holding cage, the adults became very active, pacing around the compound, lunging, and visually and olfactorily examining the holding cage doors. Upon release, the infants entered the compound single file and in constant contact ("choo-choo" formation; see Harlow, 1969). The infants walked around the compound in the choo-choo formation, followed by the adults. The infants did not immediately approach their mothers, nor did the mothers immediately run to them. The female

10982302, 1994, 7, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/dev.420270703 by Universitaetsbibliothek, Wiley Online Library on [01/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

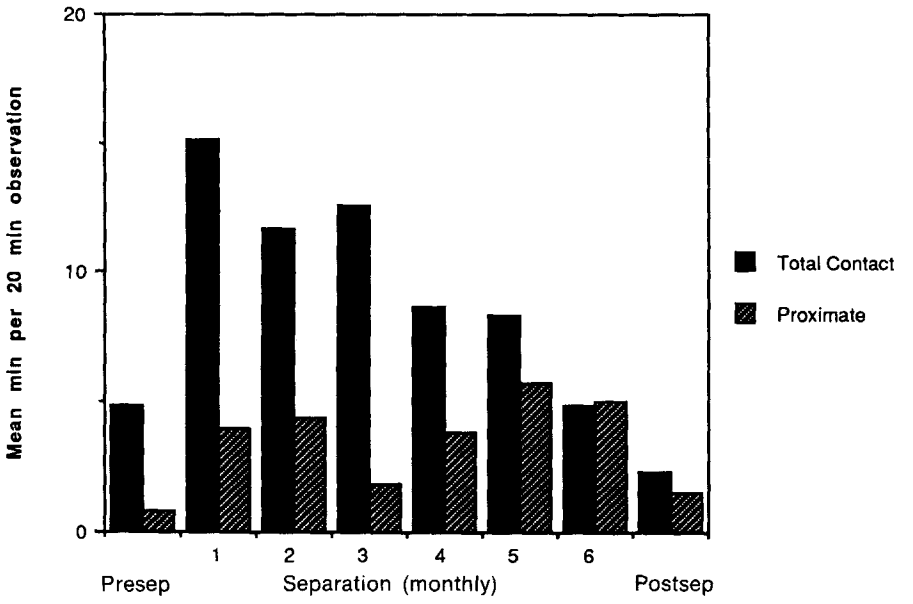



Fig. 3. Preseparation, separation (monthly), and postseparation infant–infant Total Contact and Proximate in mean minutes per 20-min observation.

infant left the group of infants 25 s later, moving to her mother and onto her back. She remained with her mother for only a few seconds, however, and then returned to the other infants. Through the next 20 min, individual infants repeated this, moving between their mothers and the other infants. The adults generally followed the infants; as an infant returned to its mother, she would take it away from the others. Approximately 20 min after reunion, the infants began playing with one another, while the adults maintained a close distance between themselves and the infants. Through the first days of reunion, infants spent more time closer to each other than to their respective mothers. Subsequent to this, infants spent more time with their mothers. For example, infants were in contact with each other for an average of 3.1 min per observation and with their mothers an average of 2.1 min in the 1st week postseparation; by the last week postseparation, infant–infant contact had declined to an average of 1.1 min per observation while mother–infant contact occurred for 2.5 min per observation.

Measures of mother–infant spatial proximity showed that mothers and infants spent more time closer together in the postseparation than preseparation periods (Table 3). Ventro–ventral, dorso–ventral, other contact (Figure 1), proximate, and distant, all occurred with a greater mean total time per 20 min with especially large increases in dorso–ventral contact (mean of 0.1 min preseparation to 1.3 min postseparation), proximate (0.1 min preseparation to 1.3 min postseparation), and distant (1.3 min to 3.3 min). Dorso–ventral was particularly high in the female infant. Follow and social examination also suggest increased mother–infant attention to one another through this final period.

Many infant–infant and infant nonsocial behaviors changed to some degree through the postseparation period. Total contact declined to below preseparation levels while proximate increased in all infants. Both forms of social play as well as object examination declined relative to preseparation while solitary play increased. Neither self-hold/fetal nor hold/hug was observed in the postseparation period.

10982302, 1994, 7, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/dev.420270703 by Universitaetsbibliothek, Wiley Online Library on [01/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

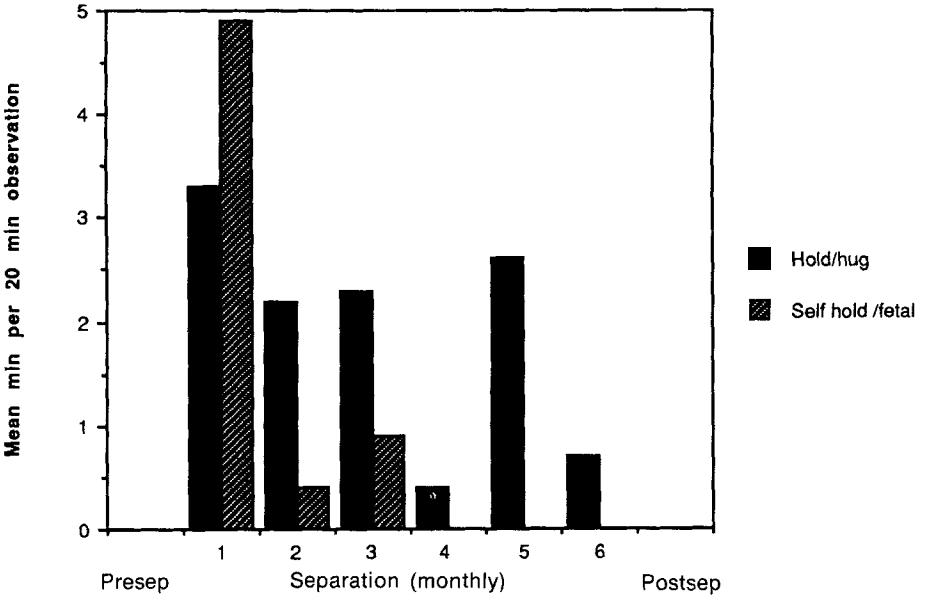



Fig. 4. Preseparation, separation (monthly), and postseparation infant–infant Hold/Hug and infant solitary Self-Hold/Fetal positions in mean minutes per 20-min observation.

## Discussion

This study was conducted to examine the effects of a lengthy separation between 3 gorilla mother–infant pairs, with the infants removed from the mothers and housed together in a small enclosure. While there were a variety of individual and social behaviors that changed in the infants through the course of separation and reunion, interpretation of the data are limited by the confound of separation and separation environment.

Through the course of separation and reunion, the infant gorillas showed a clear behavioral depression much like the anaclitic depression in children described by Bowlby (1960, 1969, 1973, 1980) as encompassing protest, despair, and detachment. Protest was characterized by loud vocalizations and increased activity for the first several days. Infants were in close proximity throughout these first days, due to increased activity levels as well as the appearance of infants holding one another. Protest gradually graded into despair, with the infants spending a good deal of each observation period huddled together in a dorso–ventral position, or alone and holding themselves. Enviromental (e.g., object examination) and peer-directed behavior (e.g., play) declined sharply. All of the above are major defining characteristics of despair as reported in humans (Bowlby, 1960, 1969, 1973, 1980; Spitz, 1946), monkeys (e.g., Hinde, Spencer-Booth, & Bruce, 1966; Kaufman & Rosenblum, 1967, 1969; Spencer-Booth & Hinde, 1967), and apes (Bard & Nadler, 1983; Codner & Nadler, 1984; Nadler & Codner, 1983). Although not specifically measured, caretakers reported that the appetites of all the infants were depressed through the first weeks of separation. Through the remaining months of separation, the infants substantially recovered from the major indices of depression. The self-hold/fetal positions and infant–infant huddling became much less common, and the time the infants were in contact was increasingly spent in play (Table 4), although active social play levels remained depressed.

10982302, 1994, 7, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/dev.420270703 by Universitaetsbibliothek, Wiley Online Library on [01/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

The separation phase reversal of active and moderate social play levels could be due to stress-related behavioral regression resulting from separation (see Harlow, Harlow, & Hansen, 1963; Seay et al., 1962). However, the nature of the separation environment suggests an alternative interpretation. The gorilla compound had a grass and soil substrate, while the separation cage was much smaller and had a concrete floor. The low incidence of active social play may be due to the fact that this floor was too uncomfortable and/or too small to engage in much rough and tumble play. Moderate social play, being much less active, could have arisen to the extent it did as a behavioral adaptation to the housing conditions. In the postseparation condition, the active and moderate social play levels returned to the preseparation relationship of active social play occurring at higher levels than moderate social play. The fact that social play occurred at much lower levels in the postseparation period than in the preseparation period was partly due to the substantial increase in mother–infant contact and the decline in infant–infant contact; the infants were with each other much less in the postseparation condition, necessarily reducing the possibilities for social play. Play as a percentage of contact (Table 4) did show a substantial rise in the postseparation condition as compared to separation, but was not as high a percentage as in the preseparation condition.

The separation cage also affected inter-infant distances. The pre- and postseparation gorilla compound was of such a size that the infants could remain as much as 38 m apart. In the separation cage, however, infants could get no more than approximately 4.3 m apart. This necessarily increased spatial proximity scores during separation. Proximity variables especially affected by this were distant, proximate, and other contact. Dorso–ventral and ventro–ventral contact were more likely to be purposeful behaviors, and therefore less influenced by cage size differences.

Recovery in these infants began while they were still separated. Kaufman and Rosenblum (1967) reported that recovery from despair in pigtail macaque infants began approximately 1 week after separation. The gorillas were slower, and perhaps less complete, in this respect, with recovery not beginning until later in the 1st month of separation. This could be a true species difference, or could be due to the different methodologies employed. Kaufman and Rosenblum removed the mothers from the infants, leaving the infants in an otherwise intact complex social group. Remaining in the group may have lessened the impact of mother–infant separation in the pigtails (see Mineka & Suomi, 1978). A third possibility, that relative ages of the gorillas versus the pigtail macaques was responsible for the difference in recovery from despair, seems less likely. The gorillas were relatively older infants (27 months) than the pigtail macaques (mean of 5.6 months). At these ages, the gorillas were spending approximately 2.5% of the observation time in any form of contact with the mothers (and no time in ventro–ventral contact), while pigtail macaques spend more than 35% of their time (see Rosenblum & Kaufman, 1967) in the intimate "ventral–ventral cling" position. These data indicate that the gorillas were relatively more independent at the time of separation and thus suggests that they would be likely to recover more, rather than less, quickly if age were the only relevant variable.

The gorilla infants showed detachment responses upon reunion with their mothers. There was no immediate ventro–ventral contact with the mother as that which characterized reunion in reports of monkeys (e.g., Kaufman & Rosenblum, 1967; Seay et al., 1962). The infants remained together initially upon reunion and spent several days in relatively close association with each other. Later in the first reunion week and following, there was a decline in infant–infant contact and an increase in mother–infant attachment behaviors. This response is similar to the detachment response in children

10982302, 1994, 7, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/dev.420270703 by Universitaetsbibliothek, Wiley Online Library on [01/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

separated from their mothers described by Bowlby (1960, 1969, 1973, 1980) as involving a disinterest or apathy and lack of attachment behaviors toward the mother upon reunion. This response is not typical of reports of most nonhuman primate species upon reunion following a mother–infant separation, in which a biphasic agitation/depression response occurs, without detachment (e.g., Hinde et al., 1966; Jensen & Tolman, 1962; Kaufman & Rosenblum, 1967; Seay et al., 1962), although more recent reports have suggested that a detachment response is possible in certain species. Codner and Nadler (1984) and Nadler and Codner (1983) separated 1 mother–infant dyad of each species of great ape for 2 weeks, and found an initial detachment upon reunion. Plimpton and Rosenblum (1981) have suggested that the reunion environment in other monkey studies, in which mothers were able to immediately retrieve infants, contributed to a lack of infant detachment responses. When mothers were restrained from retrieving infants by wire boxes, behaviors indicative of detachment were exhibited by infants (but see Mineka, 1982). In the current study, the mothers were not restrained from approaching the infants, and still did not immediately run to and retrieve the infants, but rather followed them closely as they moved around the compound together.

The detachment response occurring over several days in the current study may be due to a species difference or the extraordinary length of time over which the separation occurred. Bowlby (1960, 1969, 1973, 1980) has noted that the severity of detachment in children is highly correlated with the length of separation. The typical mother–infant separation occurs over a period of days (e.g., Reite, Kaemingk, & Boccia, 1989) or weeks (e.g., Seay et al., 1962). It is possible that the 24-week period of separation contributed to the observed detachment in the infants.

There were individual differences apparent in several of the behaviors through the separation months. The female infant (Machi) exhibited less active social play than the males throughout the study. This is consistent with the developmental pattern of play in these gorillas (Hoff et al., 1981b), as well as other reports of sex differences in play among wild gorillas (Fossey, 1979). Another difference is shown in the separation months contact and solitary play data. The male infant, Bom-Bom, showed consistently less contact with the other infants by the 2nd separation month and more solitary play than the others by the 3rd separation month, indicating a quicker recovery than either the other male or the female infant. A final difference of note is found in separation Month 5 in the hold/ behavior. The male infant, Akbar, exhibited this behavior (primarily with the female, Machi), at levels higher in this month than seen in any animal in any other month. Along with this was an increase in proximity by Bom-Bom, indicating that the infants were spending significant amounts of time close together. The majority of this increase in all of the animals was in the first 2 weeks of the 5th separation month. No reason for this short-term change could be identified (i.e., change in caretakers, feeding schedule/food, etc.).

The gorillas in the present study showed a course of depression more similar to reports in children than most studies of monkeys. Infant monkeys, when separated from their mothers, typically show a biphasic response of agitation upon separation, followed by a behavioral depression, and intensification of attachment behaviors upon reunion. The third stage, detachment, is not typically found in monkeys but was found in this study, consistent with other reports of the great apes (Codner & Nadler, 1984; Nadler & Codner, 1983). This finding supports the notion that the close biochemical relationship of the gorilla to humans (Miller, 1977) makes this species particularly useful for investigating some aspects of human behavior.

10982302, 1994, 7, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/dev.420270703 by Universitaetsbibliothek, Wiley Online Library on [01/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

## Notes

We appreciate the help and cooperation of the caretaking staff of the Yerkes Regional Primate Research Center (YRPRC). This investigation was supported in part by NIH grant #RR-00165 from the National Center for Research Resources to the YRPRC and NSF grants BNS75–06287 and BNS91–09441 to R.D.N. The YRPRC is fully accredited by the American Association of Laboratory Animal Care.

## Footnote

[1] These data were collected as part of an ongoing study of infant development and mother–infant interactions in gorillas. These data were collected and presented at a professional meeting (Hoff, Nadler, & Maple, 1979) and then filed. They were resurrected in 1992 in anticipation of modifying group membership among the gorilla groups at Zoo Atlanta. The outcome of the current study allowed us to develop a strategy for reducing the possibility of behavioral depression in some young gorillas that were to be separated from their mothers.

## References

Akiskal, H. S., & McKinney, W. T. (1975). Overview of recent research in depression. *Archives of General Psychiatry, 32,* 285–305.

Bard, K., & Nadler, R. D. (1983). The effect of peer separation in young chimpanzees (*Pan troglodytes*). *American Journal of Primatology, 5,* 25–37.

Bowlby, J. (1960). Grief and mourning in infancy and early childhood. *The Psychoanalytic Study of the Child, 15,* 9–52.

Bowlby, J. (1969). *Attachment and loss: Attachment.* New York: Basic Books.

Bowlby, J. (1973). *Separation: Anxiety and anger.* New York: Basic Books.

Bowlby, J. (1980). *Loss: Sadness and depression.* New York: Basic Books.

Codner, M. A., & Nadler, R. D. (1984). Mother–infant separation and reunion in great apes. *Primates, 25,* 204–217.

Erwin, J., Brandt, E. M., & Mitchell, G. D. (1973). Attachment formation and separation in heterosexually naive preadolescent rhesus monkeys (*Macaca mulatta*). *Developmental Psychobiology, 6,* 531–538.

Fossey, D. (1979). Development of the mountain gorilla (*Gorilla gorilla beringei*): The first thirty-six months. In D. A. Hamburg & E. R. McCown (Eds.), *The great apes* (pp. 139–184). Menlo Park, CA: Benjamin Cummings.

Goodall, J. (1990). *Through a window.* Boston: Houghton Mifflin.

Harlow, H. F. (1969). Age-mate or peer affectional system. In D. S. Lehrman, R. A. Hinde, & E. Shaw (Eds.), *Advances in the study of behavior II* (pp. 333–383). New York: Academic Press.

Harlow, H. F., Harlow, M. K., & Hansen, E. W. (1963). The maternal affectional system of rhesus monkeys. In H. L. Rheingold (Ed.), *Maternal behavior in mammals* (pp. 254–281). New York: Wiley.

Hinde, R. A. (1972). Mother–infant separation in rhesus monkeys. *Journal of Psychosomatic Research, 16,* 227–228.

Hinde, R. A., & Spencer-Booth, Y. (1971). Effects of brief separation from mother on rhesus monkeys. *Science, 173,* 111–118.

Hinde, R. A., Spencer-Booth, Y., & Bruce, M. (1966). Effects of 6-day maternal deprivation on rhesus monkey infants. *Nature, 210,* 1021–1023.

Hoff, M. P., Nadler, R. D., & Maple, T. (1979, June). *Separation and depression in infant gorillas.* Paper presented at the meeting of the Animal Behavior Society, New Orleans, LA.

Hoff, M. P., Nadler, R. D., & Maple, T. (1981a). Development of infant independence in a captive group of lowland gorillas. *Developmental Psychobiology, 14,* 251–265.

Hoff, M. P., Nadler, R. D., & Maple, T. (1981b). The development of infant play in a captive group of lowland gorillas (*Gorilla gorilla gorilla*). *American Journal of Primatology, 1,* 65–72.

Jensen, J. D., & Tolman, C. W. (1962). The effect of brief separation and mother–infant specificity. *Journal of Comparative and Physiological Psychology, 55,* 131–136.

Jones, B. C., & Clark, D. L. (1973). Mother–infant separation in squirrel monkeys living in a group. *Developmental Psychobiology, 6,* 259–269.

Jones, C., & Sabater Pí, J. (1971). *Comparative ecology of Gorilla gorilla gorilla (Savage and Wyaman) and Pan troglodytes (Blumenbach) in Rio Muni, West Africa.* New York: S. Karger.

10982302, 1994, 7, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/dev.420270703 by Universitaetsbibliothek, Wiley Online Library on [01/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Kaplan, J. (1970). The effects of separation and reunion on the behavior of mother and infant squirrel monkeys. *Developmental Psychobiology, 3,* 43–52.

Kaufman, I. C., & Rosenblum, L. A. (1967). The reaction to separation in infant monkeys: Anaclitic depression and conservation–withdrawal. *Psychosomatic Medicine, 29,* 648–675.

Kaufman, I. C., & Rosenblum, L. A. (1969). Effects of separation from mother on the emotional behavior of infant monkeys. *Annals of The New York Academy of Sciences, 159,* 681–695.

Lawick-Goodall, J. van. (1971). *In the shadow of man.* Boston: Houghton Mifflin.

Miller, D. A. (1977). Evolution of primate chromosomes. *Science, 198,* 1116–1124.

Mineka, S. (1982). Depression and helplessness in primates. In H. E. Fitzgerald, J. A. Mullins, & P. Gage (Eds.), *Studies of development in nonhuman primates* (pp. 197–242). New York: Plenum Press.

Mineka, S., & Suomi, S. J. (1978). Social separation in monkeys. *Psychological Bulletin, 85,* 1376–1400.

Nadler, R. D. (1976). Rann vs. Calabar: A study in gorilla behavior. *Yerkes Newsletter, 13,* 11–14.

Nadler, R. D., & Codner, M. A. (1983). Maternal separation and reunion of an infant orang-utan. *Primates, 24,* 67–76.

Napier, J. R., & Napier, P. H. (1967). *A handbook of living primates.* New York: Academic Press.

Novak, M. A. (1973). *Fear–attachment relationships in infant and juvenile rhesus monkeys.* Unpublished doctoral dissertation, University of Wisconsin, Madison. (cited in Suomi & Harlow, 1977).

Pavlov, J. P. (1928). *Lectures on conditioned reflexes.* New York: International Publishers.

Plimpton, E., & Rosenblum, L. A. (1981). Detachment–avoidance responses to mother following a separation: A comparative perspective. In H. F. Harlow, L. A. Rosenblum, & S. Suomi (Eds.), *Advances in the study of primate social development* (pp. 146–159). New York: Van Nostrand.

Preston, D. G., Baker, R. P., & Seay, B. (1970). Mother–infant separation in the patas monkey. *Developmental Psychology, 3,* 298–306.

Rasmussen, K. L. R., & Reite, M. (1982). Loss-induced depression in an adult macaque monkey. *American Journal of Psychiatry, 139,* 679–681.

Reite, M., Kaemingk, K., & Boccia, M. L. (1989). Maternal separation in bonnet monkey infants: Altered attachment and social support. *Child Development, 60,* 473–480.

Rosenblum, L. A., & Kaufman, I. C. (1967). Laboratory observations of early mother–infant relations in pigtail and bonnet macaques. In S. A. Altmann (Ed.), *Social communication among primates* (pp. 33–41). Chicago: University of Chicago Press.

Rosenblum, L. A., & Kaufman, I. C. (1968). Variations in infant development and response to maternal loss in monkeys. *American Journal of Orthopsychiatry, 38,* 418–426.

Schaller, G. (1963). *The mountain gorilla.* Chicago: University of Chicago Press.

Seay, B., Hansen, E., & Harlow, H. F. (1962). Mother–infant separation in monkeys. *Journal of Child Psychology and Psychiatry, 3,* 123–132.

Spencer-Booth, Y., & Hinde, R. A. (1967). The effects of separating rhesus monkey infants from their mothers for six days. *Journal of Child Psychology and Psychiatry, 7,* 179–197.

Spencer-Booth, Y., & Hinde, R. A. (1971a). Effects of brief separations from mothers during infancy on behaviour of rhesus monkeys 6–24 months later. *Journal of Child Psychology and Psychiatry, 12,* 157–172.

Spencer-Booth, Y., & Hinde, R. A. (1971b). The effects of 13 days maternal separation on infant rhesus monkeys compared with those of shorter and repeated separations. *Animal Behaviour, 19,* 595–605.

Spitz, R. A. (1946). Anaclitic depression. *The Psychoanalytic Study of the Child, 2,* 313–347.

Suomi, S. J., Eisele, C. D., Grady, S. A., & Harlow, H. F. (1975). Depression in adult monkeys following separation from nuclear family environment. *Journal of Abnormal Psychology, 84,* 576–578.

Suomi, S. J. & Harlow, H. F. (1977). Production and alleviation of depressive behaviors in monkeys. In J. Maser & M. E. P. Seligman (Eds.), *Psychopathology: Experimental models* (pp. 131–173). San Francisco: Freeman.

Suomi, S. J., Harlow, H. F., & Domek, C. J. (1970). Effect of repetitive infant–infant separation of young monkeys. *Journal of Abnormal Psychology, 76,* 161–172.

Tilford, B. L., & Nadler, R. D. (1978). Male parental behavior in a captive group of lowland gorillas, (*Gorilla gorilla gorilla*). *Folia Primatologica, 29,* 218–228.

10982302, 1994, 7, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/dev.420270703 by Universiteitsbibliotheek, Wiley Online Library on [01/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License